



NEW PRODUCTS

MAKE IN INDIA

COMPONENTS

Single-chip balun

STMicroelectronics has introduced a matched balun for its S2-LP 868-927MHz low-power radio transceiver that helps engineers save board space and minimise

RF circuit design challenges in size- and cost-conscious products like the Internet of Things (IoT) sensors, smart meters, alarms, remotes, building automation and industrial controls. The 3.26mm² BALF-SPI2-01D3 integrates all the impedance-matching and filtering components needed to connect an antenna to the S2-LP radio. Thus it replaces a conventional network of 16 discrete capacitors and inductors that can occupy up to 100mm² of board real-estate, achieving a footprint reduction of more than 96 per cent.

STMicroelectronics
www.st.com

Image sensor platform

ON Semiconductor has announced a scalable family of CMOS image sensors for advanced driver assistance systems (ADAS) and autonomous driving applications. The AR0820AT, AR0220AT and AR0138AT sensors have resolutions scaling from 8.3 megapixels (MP) down to 1.2MP with low-light performance from 4.2µm pixels. Additional features include functional safety up to ASIL-C, high dynamic range (HDR), an industry-first cybersecurity option, and second-generation wafer stacking technology that reduces package size.

ON Semiconductor
<http://www.onsemi.com>

DC-DC controller

Analog Devices, Inc. has announced the Power by Linear LTC7821, a hybrid

step-down synchronous controller that merges a switched capacitor circuit with a synchronous step-down controller, enabling up to 50 per cent reduction in DC/DC converter solution size compared to traditional step-down solutions. This improvement is enabled by a three times higher switching frequency without compromising efficiency. Alternatively, when operating at the same frequency, a LTC7821-based solution can provide up to 3 per cent higher efficiency.

Analog Devices, Inc.
www.linear.com/product/LTC7821

DLP chipset

Texas Instruments (TI) has unveiled a state-of-the-art DLP chipset that combines full programmability and a resolution of more than one million addressable pixels per headlight—exceeding the resolution of existing adaptive driving beam (ADB) technologies by more than 10,000 times.

Automakers and Tier-1 suppliers can use this new programmable ADB solution to design headlight systems that maximise brightness for drivers on the road while minimising the glare to oncoming traffic or reflections from retroreflective traffic signs.

Texas Instruments
www.ti.com

Smart sensor data-acquisition device

Automotive OEMs can enhance the safety of their lithium-ion battery packs with robust communications, comprehensive diagnostics and lower system costs enabled by Maxim Integrated's MAX17843 12-channel, high-voltage

smart sensor data-acquisition device.

MAX17843 meets high safety standards, adhering to ISO 26262 and ASIL D requirements. Its differential universal asynchronous receiver/transmitter (UART) using capacitive isolation reduces bill-of-materials (BOM) costs and failure-in-time (FIT) rates. With this, designers can save up to 90 per cent of their isolation BOM cost.

Maxim Integrated
www.maximintegrated.com/MAX17843



TEST & MEASUREMENT

High-voltage probe

Kusam-Meco has introduced a high-voltage probe (Model-PD28) that can be used to measure both AC and DC voltages in conjunction with a digital multimeter. The probe extends the measuring range of digital multimeters for low-power, high-voltage, high-impedance circuits such as TV sets and video monitors.

The probe consists of two kinds of contact tips. One is a round needle tip for normal use and the other is a flat spring type for CRT anode. In addition, there is a test clip for the ground, a set of test leads for connecting to a multimeter and a high-quality enclosure with very high insulation properties. The accuracy is $\pm 1\%$ (0-20kV DC), $\pm 2\%$ (20-40kV DC) and $\pm 5\%$ (0-28kV AC).

Kusam Electrical Industries
www.kusamelectrical.com



Industrial imaging multimeter

FLIR DM285 is an industrial true-RMS digital multimeter and thermal imager. Featuring infrared guided measurement (IGM) powered by a built-in thermal imager, the meter visually guides users to the precise location of a problem such as hot spot. DM285 features Bluetooth, which enables wireless data transfer by